

Could trees make Gozo net zero?

The afforestation in the Nationalist Party's 2026 pledge, tested on measured forests and Gozo's land

THE CLAIM UNDER REVIEW

“By 2040 Gozo becomes a Net-Zero Island ... where every unit of emissions generated is reduced or offset”, with “afforestation” among the initiatives.

Partit Nazzjonalista, programme Nifs Ġdid (May 2026), chapter Ghawdex, item 46 (our translation).

News summaries said “through afforestation” and reported an indigenous-tree strategy; neither is the PN's text.

PLEDGE: NOT MEASURABLE

As of 10 Oct 2026: no area, species, rate or date for afforestation

Version 1.0 · 10 October 2026

Status: draft, pending right of reply

Prepared from public sources and open data. No site visits.

Repository: github.com/leandergrech/Mizien

MIŻIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

Before the general election of 30 May 2026 [22], the **Nationalist Party** promised a plan to make **Gozo a Net-Zero Island by 2040**, “where every unit of emissions generated is reduced or offset”, listing **afforestation** among the initiatives [1]. News reports said net zero “through afforestation” [4] and reported an indigenous-tree strategy [5 ♦]. Claim Check 107 tested the net-zero plan; this check tests what new forest could do on Gozo.

- 1 One item in a list, with no numbers.** Afforestation appears once in the Gozo chapter and twice in the environment chapter, including “more afforestation” in Malta and Gozo [1, 2], with no area, species, sites, carbon rate or date.
- 2 Trees alone could not do it.** Offsetting Gozo’s energy CO₂ (118,000–154,000 t a year, 2016–2020 [6]) would take new forest 4.9–6.3 times Gozo’s area at a rate measured in a dry pine forest [7], and 2.3–25 times across published rates [8, 9].
- 3 Realistically, a few per cent.** CORINE maps no forest on Gozo. Planting all of its 1,270 ha of garrigue, maquis and sparse vegetation [18] would offset about 3–4% of that CO₂ (1–8% across rates), and over a third of that land lies in Natura 2000 sites [23].
- 4 Water and time limit it further.** Malta is in permanent water scarcity [17]; 30–58% of planted pines survived 20 years in an arid Spanish trial [12]; native oak plantations in Spain added no soil carbon in up to 24 years [8].
- 5 No indigenous-tree strategy in the programme. Pledge label: Not measurable (10 October 2026).** “Indigenous” appears once, for farm varieties and breeds [3]. A pledge is not false; as worded, its afforestation cannot be shown to be delivered or missed.

THIS CLAIM

NOT
MEASURABLE

NOT YET
DUE

ON
TRACK

OFF
TRACK

MET

MISSED

4.9–6.3×

Gozo’s area needed as new forest to offset its energy CO₂ (measured rate)

3–4%

of that CO₂ offset if all Gozo’s semi-natural land were planted

0 ha

of forest mapped on Gozo (CORINE 2018)

Not found

an indigenous-tree strategy in the 16 programme chapters

WHAT WOULD MAKE IT MEASURABLE

An afforestation target for Gozo with area, sites, species, years and the carbon rate assumed, within a Gozo emissions inventory and a 2040 pathway.

WHAT WOULD COUNT AGAINST IT

A plan that counts on Gozo’s trees for more than a few per cent of its emissions, or plants on garrigue the programme itself would protect.

IN THIS REPORT

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4 What the studies found

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1 The pledge and what we could verify

Our candidate record took the claim from a MaltaToday breakdown of the programme [5], which says Gozo would become net zero through afforestation alongside a national indigenous-tree strategy. That article could not be read (HTTP 403; the Internet Archive was unreachable), so we went to the programme itself, *Nifs Ġdid*, published by the PN as 16 web chapters and 16 chapter PDFs (the pages are dated 18 May 2026). We read all of them on 10 October 2026 and searched each for the words this check turns on; the counts and file hashes are in *data/cc-092/pn_programme_search.csv*. The programme is in Maltese; translations are ours.

Source	What it says (our translation)	Our access	Status
PN, <i>Nifs Ġdid</i> , chapter Ghawdex, item 46 [1]	A plan so that by 2040 Gozo becomes a Net-Zero Island, “where every unit of emissions generated is reduced or offset” (<i>imnaqqsa jew ikkumpensata</i>) by initiatives such as solar panels, electric vehicles, building efficiency, ecological restoration on land and at sea, “afforestation” (<i>afforestazzjoni</i>) and water and soil conservation, “among others”.	Read in full (web page and PDF p. 16).	The pledge
PN, chapter Ambjent, items 01, 20 and 25 [2]	01: a park in every locality; “the plan also includes more afforestation and open spaces in both Malta and Gozo”. 20: tree planting and afforestation, with valley restoration and rain gardens, “to fight urban heat, flooding and pollution”. 25: a Malta Nature Network protecting valleys, rubble walls and garrigue (<i>xagħir</i>).	Read in full (PDF pp. 8, 12, 13).	Related pledges
PN, chapter Agrikoltura u Sajd, item 31 [3]	Conservation and propagation of local plant varieties and livestock breeds (<i>indigeni</i>), with a gene bank, nursery and support programme within three years.	Read in full (PDF p. 13).	Only use of “indigenous”
MaltaToday, May 2026 [4]	Gozo would become a net-zero island through afforestation.	Read via the Internet Archive for Claim Check 107 (5 Oct 2026).	News summary
MaltaToday, breakdown, May 2026 [5 ♦]	Seen only as a search engine’s summary: a nationwide afforestation strategy prioritising indigenous trees and valley restoration.	Not read (403).	Locator; second-hand

FAIRNESS NOTE

Labour won the general election of 30 May 2026 [22], so this is an opposition proposal. The Government’s own aim of a climate-neutral Gozo (Claim Check 031) and Labour’s 2022 pledge of 100,000 trees (Claim Check 010) are held to the same standard. The PN’s environment chapter presents trees mainly as a defence against heat, flooding and pollution [2]; those benefits are real but are not what this check measures. Nothing here says afforestation is undesirable or that anyone acted in bad faith.

2 Method

Question. As worded, can the afforestation in the PN’s pledge be measured, and how much of Gozo’s emissions could new forest on Gozo offset? Does the programme contain an indigenous-tree strategy?

Gozo’s emissions. There is no official greenhouse-gas inventory for Gozo. We use the energy-related CO₂ estimated for 2016–2020 in the 2023 Energy Baseline Scenario for Gozo [6] (118,333–153,997 t a year; electricity, transport on the island, ferries, heating and cooling), as in Claim Check 107, and as a cross-check Malta’s 2024 inventory total scaled by Gozo’s population share (157,634 t CO₂e [17]), as in Claim Check 031.

Sequestration rates. Three cases from the literature, each verified in Crossref: *low*, 0.92 t CO₂ per hectare a year, the lower end of tree-biomass gains in plantations of native holm and cork oak in Spain [8]; *central*, 3.62 t, the 35-year gain in trees and soil in the Yatir Aleppo pine forest in Israel [7]; *high*, 7.6 t, a global review’s rate for planted pine in dry temperate climates over the first 20 years [9].

Land and water. Gozo’s land cover from CORINE Land Cover 2018 [18]; land areas from the European Commission (Gozo 67 km² [20]) and Eurostat (Gozo and Comino 68 km² [17]); Malta’s forest area, forest-land removals and water exploitation index from Eurostat [17]; Natura 2000 sites from the EEA [23]; rainfall from NASA POWER [19]; survival and water use from field and review studies [11–13].

Grades. A experiment; B observational study with a control; C review, guidance or official statistics; D assertion. Our arithmetic gives ceilings, not forecasts. Code: *tools/cc-092-report/calc.py*; every figure in *data/cc-092/checks.csv*.

3 How new forests store carbon

A new forest takes CO₂ from the air as its trees grow and stores the carbon in wood and roots and in the soil [7, 9]. The rate varies with climate, site, species and management [9], and with what the land held before: in the Yatir forest, half the measured gain was in the soil [7], while native oaks planted on former cropland in Spain gained carbon in their wood but not in the soil [8]. The review rates used here count tree biomass only [9]. Trees must also survive: in an arid Spanish trial, 30–58% of planted pines were alive after 20 years [12]. Figure 1 sets the measured rates against what Gozo would need: to offset its energy CO₂ on its own 6,700 ha, every hectare would have to store 17.7–23.0 t a year.

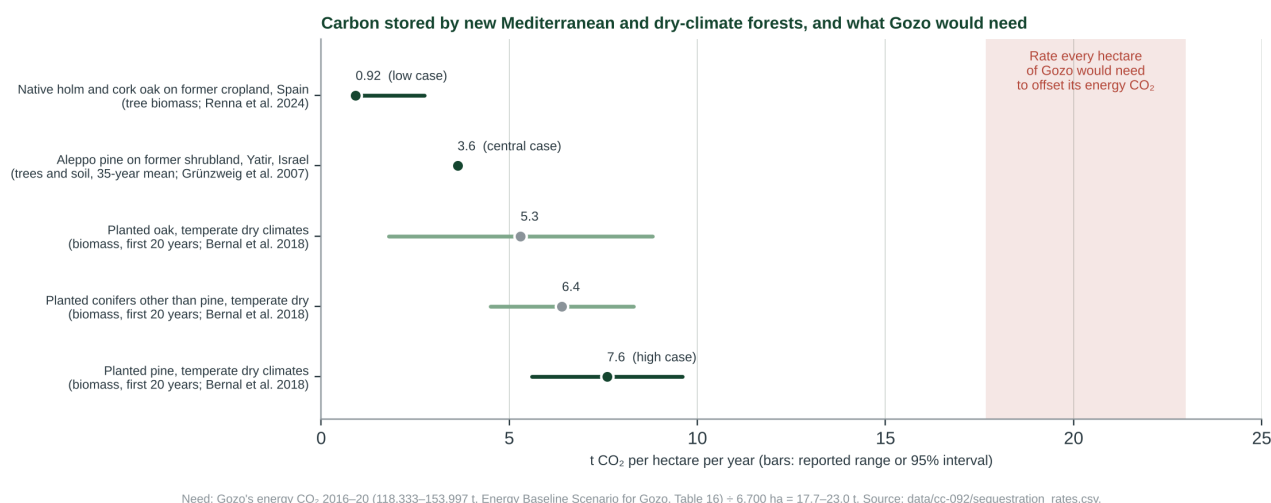


Figure 1. Carbon stored by new forests in Mediterranean and dry climates (t CO₂ per hectare a year), and the rate every hectare of Gozo would need to offset its 2016–2020 energy CO₂. Bars: the reported range [8] or the 95% interval [9].

4

What the studies found

Study	Design and place	Finding used here	Grade
Grünzweig et al. 2007 [7]	Aleppo pine planted on grazed shrubland, Yatir, Israel; 35 years; compared with the shrubland. Not irrigated.	Ecosystem carbon rose by 99 g C/m ² a year (3.62 t CO ₂ /ha), half of it in soil.	B
Renna et al. 2024 [8]	Native holm and cork oak on former cropland, Extremadura, Spain; plots 11–24 years old against unplanted controls.	Tree biomass 25–75 g C/m ² a year (0.92–2.75 t CO ₂ /ha); soil carbon not higher, so total carbon was not significantly higher; may take decades to become a sink.	B
Bernal et al. 2018 [9]	Global synthesis of more than 335 studies and reports; plantation growth by species and climate.	Planted forests, dry temperate climates, first 20 years: oak 5.3, other conifers 6.4, pine 7.6 t CO ₂ /ha a year (biomass only).	C
Hoogmoed et al. 2012 [10]	Meta-analysis of pasture afforestation in Mediterranean climates.	No substantial change in soil carbon across three decades.	C
Rotenberg and Yakir 2010 [14]	Nine-year field study at the forests' dry timberline.	Forests where 200–600 mm of rain falls keep sequestering, but need several decades of carbon gain to balance the warming from their darker surface.	B
Oliet et al. 2023 [12]	Field experiment, Aleppo pine, arid south-eastern Spain, 20 years.	Survival 57.5% with mesh shelters, 46% unprotected, 29.5% in tube shelters.	A
Maestre and Cortina 2004 [11]	Review of Aleppo pine plantations in semi-arid Mediterranean areas.	Higher water use, often more runoff and soil loss than shrubland, mostly negative effects on native vegetation.	C
Jackson et al. 2005 [13]	Synthesis of more than 600 observations, with modelling.	Plantations cut stream flow by 52% (227 mm a year) on average.	C
Veldman et al. 2015 [16]	Perspective on tree planting in open ecosystems.	Planting forest on grasslands and open woodlands harms biodiversity and ecosystem services.	C

No measurement of afforestation carbon in Malta or Gozo was found (Crossref searches, 10 October 2026). The rates come from Spain, Israel and global syntheses; Gozo's rainfall, about 570 mm a year in a reanalysis [19], is higher than Yatir's (about 300 mm [15]).

5

Where the science disagrees

Does afforestation in dry Mediterranean climates add soil carbon?		UNSETTLED
EVIDENCE OF A SOIL GAIN At Yatir, soil organic carbon rose by 50 g C/m ² a year over 35 years after pine was planted on grazed shrubland, about half of the forest's total gain [7].	EVIDENCE OF LITTLE OR NO SOIL GAIN Native oaks planted on former cropland in Spain added no soil carbon after up to 24 years [8]; a meta-analysis of pasture afforestation in Mediterranean climates found no substantial change across three decades [10].	
WHY THEY DIFFER, AND WHAT IT MEANS HERE Former land use (grazed shrubland against cropland or pasture), species, age and soils differ. For Gozo we let the central case keep Yatir's soil gain and the low case have none. Either way the forest area needed is several times Gozo's size.		

How soon does a new dry-land forest cool the climate?

DEPENDS ON SCALE

FASTER BENEFIT

In a climate model, semi-arid afforestation on the scale of about 200 million hectares changed rainfall and would outweigh its surface warming within about six years [15].

SLOWER BENEFIT

Measured over nine years at the forests' dry timberline, the darker forest surface warms the climate enough that several decades of carbon gain are needed to balance it [14].

WHY THEY DIFFER, AND WHAT IT MEANS HERE

One is a field study at the dry edge of forests, the other a model of a continent-sized planting. For a small island and a 2040 deadline, the field result is the closer guide: carbon accounting alone would overstate what young trees do for the climate by 2040. We do not adjust our figures for this.

Can new trees establish on Gozo without irrigation?

POSSIBLE, WITH LOSSES

EVIDENCE TREES CAN ESTABLISH

Yatir grew without irrigation on about 300 mm of rain [7, 15], and dry forests keep sequestering where 200–600 mm falls [14]; Gozo receives about 570 mm a year (reanalysis) [19].

EVIDENCE OF LIMITS

In arid Spain 42–70% of planted pines died within 20 years [12]; plantations use more water and reduce stream flow [11, 13]; Eurostat records Malta in permanent water scarcity, with a water exploitation index of 30% in 2023 (above 20% signals scarcity) [17].

WHY THEY DIFFER, AND WHAT IT MEANS HERE

Rainfall is not the same as water available to a seedling: soil depth, drought years and planting technique decide survival. Trees can grow on Gozo, but survival and water use limit how much planting adds, and the PN does not say how plantations would be watered.

6

The arithmetic

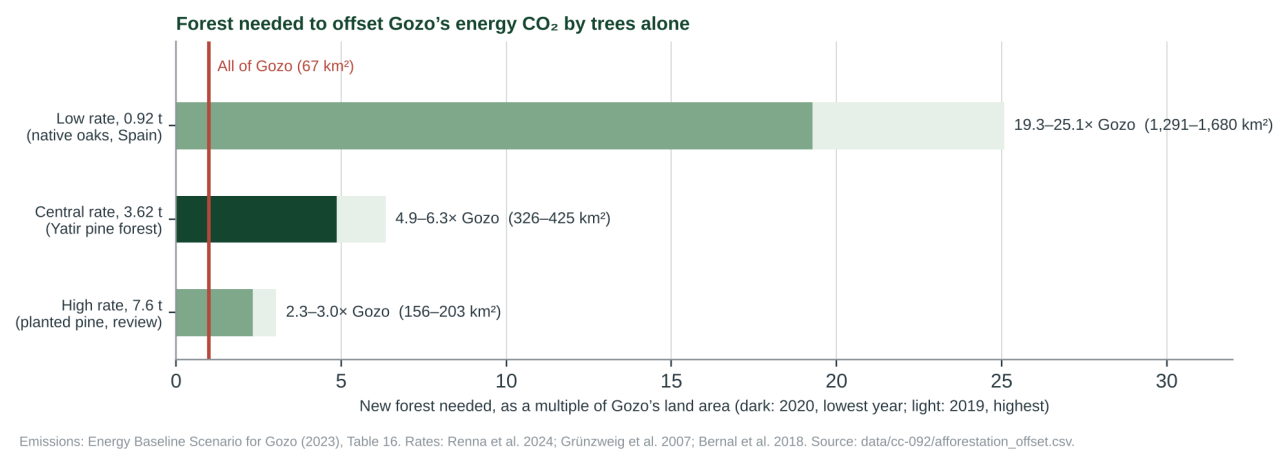


Figure 2. New forest needed to offset Gozo's 2016–2020 energy CO₂ [6] by trees alone, as a multiple of Gozo's 67 km² [20], at the three rates.

Rate (t CO ₂ /ha a year)	Forest needed for 118,333 t (2020)	Forest needed for 153,997 t (2019)	Multiple of Gozo
Low, 0.92 (native oaks [8])	1,291 km ²	1,680 km ²	19.3–25.1×
Central, 3.62 (Yatir [7])	326 km ²	425 km ²	4.9–6.3×
High, 7.6 (planted pine [9])	156 km ²	203 km ²	2.3–3.0×

With Malta's 2024 inventory scaled by Gozo's population share (157,634 t CO₂e [17]) the central case needs 435 km², 6.5 times Gozo. Full results: data/cc-092/afforestation_offset.csv.

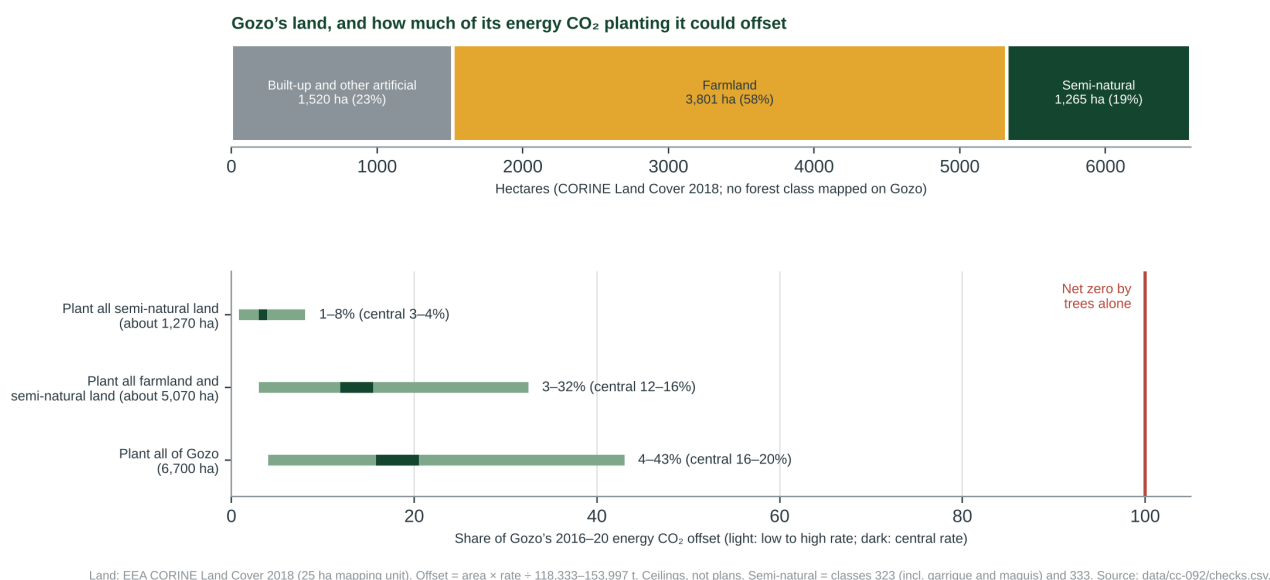


Figure 3. Gozo's land in CORINE Land Cover 2018 [18] (6,586 ha mapped), and the share of its 2016–2020 energy CO₂ that new forest on that land could offset. These are ceilings: they assume every hectare is planted and reaches the rate.

THE ARITHMETIC

In CORINE, Gozo's only land that is neither built on nor farmed is about 1,270 ha (19%): class 323, sclerophyllous vegetation, which includes garrigue and maquis [18], and class 333, sparsely vegetated areas. Planted in full at Yatir's rate, it would store $1,265 \text{ ha} \times 3.62 \text{ t} = \text{about } 4,600 \text{ t CO}_2$ a year, 3.0–3.9% of Gozo's energy CO₂ (0.8–8.1% across the three rates). Planting all farmland too (5,066 ha) would reach 12–16%; covering all of Gozo, 16–21%. Malta's whole forest is 470 ha (FAO definition, 2025) [17], and in 2024 Malta's forest land removed 0.16 kt CO₂e [17], 0.1% of Gozo's 2019 energy CO₂.

What this means for the pledge

- **Afforestation can be a small part of a net-zero Gozo, not the means to it.** The PN's text relies on cutting emissions first (solar, electric vehicles, efficient buildings) and lists afforestation among several measures [1]; that is consistent with this arithmetic. The “through afforestation” version is a news summary [4].
- **The planting effort would be large even for a few per cent.** Stocking the semi-natural land at Yatir's 300 trees per hectare [7], allowing for 46% survival after 20 years [12], means about 825,000 trees, about 52 years at the pace at which government entities planted trees across Malta and Gozo in 2022–2025 (about 60,000 trees [21]).
- **That land is not empty.** Class 323 includes garrigue, which the PN's own Malta Nature Network (item 25) would protect [2], and 36% of Gozo's semi-natural land (453 ha) lies inside Natura 2000 sites [23]. Planting trees in open habitats can harm biodiversity [16], and Aleppo pine plantations have often suppressed native vegetation [11].

7 Testing the pledge

Sub-claim	Said by	What the evidence shows	Rating
A. Every unit of Gozo's emissions "reduced or offset" by 2040, with afforestation among the initiatives	PN [1]	The PN gives afforestation no share. Trees alone would need 4.9–6.3 times Gozo's area at a measured rate (2.3–25 times across rates); planting all of Gozo's semi-natural land would offset about 3–4% (1–8%) of its energy CO ₂ [6–9, 18].	A SMALL PART AT MOST
B. "More afforestation and open spaces in both Malta and Gozo"	PN [2]	No area, number of trees, species, sites or date, so delivery cannot be checked. CORINE maps no forest on Gozo; Malta's whole forest is 470 ha [17, 18].	NOT MEASURABLE (PLEDGE)
C. Gozo net zero "through afforestation"	MaltaToday [4]	Not the PN's words: the programme lists afforestation among many measures [1]. Not rated.	HEADLINE WORDING, NOT RATED
D. A national indigenous-tree strategy	MaltaToday [5 ♦]	Not in any of the 16 chapters (web and PDF, searched 10 Oct 2026). Nearest: local farm varieties and breeds with a gene bank and nursery (Agrikultura 31) [3]; tree planting and afforestation against heat and flooding (Ambjent 20) [2].	NOT IN THE PROGRAMME
E. Water and land limits on planting (our check)	Not claimed	Malta in permanent water scarcity (WEI+ 30% in 2023) [17]; 30–58% 20-year survival in arid Spain [12]; the only land neither built on nor farmed is mostly CORINE class 323, which includes garrigue and maquis [18]; 36% of it lies in Natura 2000 sites [23], and the PN's item 25 lists garrigue among the habitats to protect [2].	CONTEXT

8 Pledge label and requests for evidence

PLEDGE

NOT MEASURABLE

As of 10 October 2026. Afforestation is listed without area, species, carbon rate or date; trees on Gozo could offset only a few per cent of its emissions.

Why. The PN lists afforestation as one of the ways Gozo's emissions would be "reduced or offset" by 2040 and promises "more afforestation" in Malta and Gozo, but sets no quantity, place or date against which delivery could be checked. That is the definition of *Not measurable* (Appendix A). The arithmetic adds context, not a verdict: whatever is planted, trees on Gozo can carry only a small share of a net-zero target. The indigenous-tree strategy reported in the press is not in the programme, so it is not rated.

What this label does not say. It does not say the pledge is false, that afforestation is a bad idea, or that anyone acted in bad faith. A checkable version would read: "By [year] we will plant [area] ha of [species] at [sites] on Gozo, expected to store [X] t CO₂ a year by 2040, within a net-zero pathway that cuts emissions to [Y] t."

Evidence we are asking for

- ☐ From the PN: the share of Gozo's emissions that afforestation is expected to offset by 2040, and the rate per hectare assumed.
- ☐ From the PN: the area, sites and species to be planted on Gozo, and whether garrigue and other open habitats would be excluded.
- ☐ From the PN: whether it proposes a national indigenous-tree strategy, and where it is set out.
- ☐ From the PN: how new plantations would be watered and their survival monitored.

RIGHT OF REPLY

Under the project's rules a pledge labelled Not measurable is offered to its author for reply before any wider circulation. The maintainer handles the right of reply to the Partit Nazzjonalista; any response will be appended and the label revisited.

9

Limitations

- No measurement of afforestation carbon in Malta was found. The three rates come from Spain, Israel and a global review; Gozo's soils, species and rainfall differ, and a review's "dry temperate" class is not a Mediterranean-only figure [9].
- Gozo's emissions come from a study prepared for the EU islands secretariat (energy only; not official statistics) and from a population share of Malta's inventory, which is not Gozo data [6, 17].
- CORINE maps land in units of at least 25 ha, so small woods, tree rows and valley vegetation are missed; polygons were assigned to Gozo by their extent. CORINE maps 6,586 ha on Gozo, against 67 km² from the Commission [20] and 68 km² for Gozo and Comino from Eurostat [17].
- The offsets are ceilings: they assume every hectare is planted and reaches the rate, before losses to drought, fire or failed planting. The Natura 2000 overlap is measured on generalised CORINE polygons and is approximate.
- Rainfall comes from a reanalysis with grid cells of about 50 km and is approximate [19].
- MaltaToday's breakdown [5] could not be read; its indigenous-tree wording is known only from a search engine's summary. The translations of the programme are ours.

References

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- [3] Partit Nazzjonalista (May 2026). *Nifs Ġdid*, chapter Agrikoltura u Sajd, item 31 (PDF 14-Agrikoltura.pdf, p. 13). Read 10 Oct 2026. <https://pn.org.mt/nifsgdid/agrikoltura-u-sajd/>
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Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Pledge labels

A pledge is a promise of future action, so it cannot be true or false when it is made. It gets one of these labels instead of a verdict, with the date of the evidence behind it.

Pledge label	Meaning
NOT MEASURABLE	The pledge as worded has no definition, baseline or date against which delivery could be checked.
NOT YET DUE	Measurable; the deadline or term has not passed and no progress data have been published yet.
ON TRACK	Published progress is at or ahead of a straight-line path (or the pledge's own milestones) to the target by the deadline.
OFF TRACK	Published progress is behind that path, and the target has not yet been shown to be met or missed.
MET	Documents or data show the target was reached, by the deadline if one was set.
MISSED	The deadline or term has passed, and documents or data that can be shown say the target was not reached.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand. Our arithmetic gives ceilings, not forecasts.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.

- **Offer a right of reply** where a check finds a claim Not substantiated, Misleading or Contradicted (or a pledge Not measurable, Off track or Missed), before wider circulation, and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	10 Oct 2026	First issue. Pledge label Not measurable as of 10 October 2026; pending right of reply from the Partit Nazzjonalista. Complements Claim Check 107 (the net-zero plan) with the afforestation arithmetic, Gozo’s land cover, water limits and the search for an indigenous-tree strategy.